

AssignmentOne

DATA TYPES & ARRAYS VERIFICATION COURSE

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1. Code Snippets

```
1  module AssignmentOne;
2
3      //===== Part_A =====
4
5      logic signed [3:0] a;
6      byte unsigned b;
7      shortint c;
8      bit [3:0] d;
9      real r;
10     string myString = "SV Assignment1";
11
12     //===== Part_B =====
13
14     //=====3=====
15
16     int OneD_unpacked_array [4:0] = '{1, 2, 3, 4, 5};
17     logic [2:0] [1:0] TwoD_packed_array = '{2'b10,2'b10,2'b10};
18     byte OneD_unpacked_array_4 [3:0] = '{10, 20, 30, 40};
19
20     //=====4=====
21
22     typedef struct {
23         string name;
24         int id;
25         byte grade;
26     } student_t;
27
28     student_t students[0:2];
29
30     //=====5=====
31
32     typedef union {
33         int i;
34         byte b;
35     } data_u;
36
37     data_u data;
38
39     //===== Part_C =====
40
41     int dyn_arr[],dyn_copy[];
```

```

1  initial begin
2      $display("===== Display Part_A =====");
3
4      a = -3;
5      b = 250;
6      c = -12345;
7      d = 4'b1101;
8      r = 3.14159;
9
10     $display("\nValue of a: %0d", a);
11     $display("\nValue of b: %0d", b);
12     $display("\nValue of c: %0d", c);
13     #5;
14     $display("\nValue of d: %0b", d);
15     #15;
16     $display("\nValue of r: %f", r);
17     $display("\nString value: %s", myString);

```

```

1  $display("\n===== Display Part_B =====");
2
3  $display("\n===== 3 =====");
4
5  $display("\nOneD_unpacked_array: %p", OneD_unpacked_array);
6  $display("\nTwoD_packed_array: %p", TwoD_packed_array);
7  $display("\nOneD_unpacked_array_4: %p", OneD_unpacked_array_4);
8
9  $display("\n===== 4 =====");
10
11  students[0] = '{name: "Ali", id: 1, grade: 85};
12  students[1] = '{name: "Sara", id: 2, grade: 90};
13  students[2] = '{name: "Omar", id: 3, grade: 78};
14
15  for (int i = 0; i < 3; i++) begin
16      $display("\nName: %s, ID:
17  endrade:
18  ", students[i].name, students[i].id, students[i].grade);
19  $display("\n===== 5 =====");
20
21  data.i = 32'h12345678;
22  $display("\nData.i: %h, Data.b: %h", data.i, data.b);
23  data.i = 8'h12;
24  $display("\nData.i: %h, Data.b: %h", data.i, data.b);

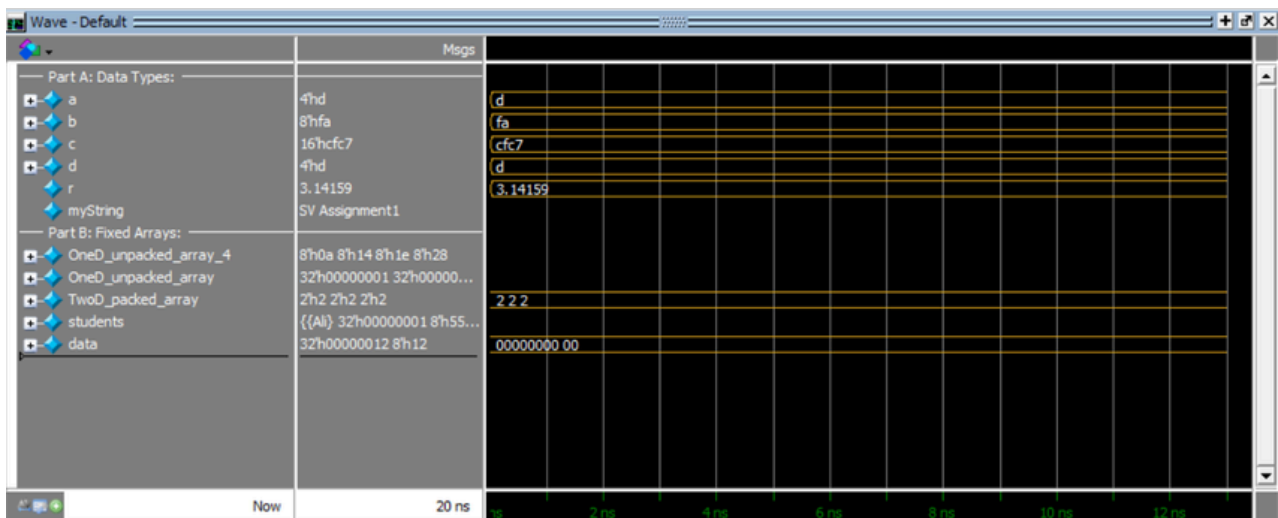
```

```

1      $display("\n===== Display Part_C =====");
2
3      dyn_arr = '{5, 10, 15, 20, 25}';
4      dyn_copy = dyn_arr;
5
6      $display("\ndyn_arr: %p, dyn_copy: %p", dyn_arr, dyn_copy);
7
8      dyn_arr = new[20](dyn_arr);
9      for(int i = 5; i < 20; i++) begin
10         dyn_arr[i] = 'hC';
11     end
12
13     $display("\ndyn_arr updated: %p", dyn_arr);
14
15     for(int i=0; i<dyn_arr.size(); i++) begin
16         $display("\ndyn_arr[
17 ]: %d", dyn_arr[i]);
18
19         foreach(dyn_copy[i]) begin
20             $display("\ndyn_copy[
21 ]: %d", dyn_copy[i]);
22
23             dyn_copy.delete();
24             $display("\ndyn_copy size after delete:
25 ", dyn_copy.size());
26
27         end
28
29     endmodule

```

2. Snippets for the waveform



3. Snippets for the logs

Part A

```
# ----- Display Part_A -----
#
# Value of a: -3
# Value of b: 250
# Value of c: -12345
# Value of d: 1101
# Value of r: 3.141590
# String value: SV Assignment1
#
```

Part B

```
# ----- Display Part_B -----
#
# ----- 3 -----
# OneD_unpacked_array: '{1, 2, 3, 4, 5}'
# TwoD_packed_array: '{2, 2, 2}'
# OneD_unpacked_array_4: '{10, 20, 30, 40}'
#
# ----- 4 -----
# Name: Ali, ID: 1, Grade: 85
# Name: Sara, ID: 2, Grade: 90
# Name: Omar, ID: 3, Grade: 78
#
# ----- 5 -----
# Data.i: 12345678, Data.b: 78
# Data.i: 00000012, Data.b: 12
```

Part C

```
# ===== Display Part_C =====
#
# dyn_arr: '{5, 10, 15, 20, 25}', dyn_copy: '{5, 10, 15, 20, 25}'
#
# dyn_arr updated: '{5, 10, 15, 20, 25, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12}'
#
# dyn_arr[0]:      5
#
# dyn_arr[1]:      10
#
# dyn_arr[2]:      15
#
# dyn_arr[3]:      20
#
# dyn_arr[4]:      25
#
# dyn_arr[5]:      12
#
# dyn_arr[6]:      12
#
# dyn_arr[7]:      12
#
# dyn_arr[8]:      12
#
# dyn_arr[9]:      12
#
# dyn_arr[10]:     12
#
# dyn_arr[11]:     12
#
# dyn_arr[12]:     12
#
# dyn_arr[13]:     12
#
# dyn_arr[14]:     12
```

```
#
# dyn_arr[15]:     12
#
# dyn_arr[16]:     12
#
# dyn_arr[17]:     12
#
# dyn_arr[18]:     12
#
# dyn_arr[19]:     12
#
# dyn_copy[0]:      5
#
# dyn_copy[1]:      10
#
# dyn_copy[2]:      15
#
# dyn_copy[3]:      20
#
# dyn_copy[4]:      25
#
# dyn_copy size after delete: 0
#
# Assignment 1 completed successfully.
```